

5.4.

$$3. \bar{X} \sim N(100, \frac{4}{15}) \quad \bar{Y} \sim N(100, \frac{1}{5})$$

$$\therefore \bar{X} - \bar{Y} \sim N(0, \frac{7}{15})$$

$$\therefore P(|\bar{X} - \bar{Y}| > 0.2) = 2P(\bar{X} - \bar{Y} > 0.2) \approx 0.7718$$

$$8. p_z(z) = \frac{\Gamma(\frac{m+n}{2}) \left(\frac{n}{m}\right)^{\frac{n}{2}}}{\Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} \cdot \left(\frac{mz}{n(1-z)}\right)^{\frac{n}{2}-1} \cdot \left(1 + \frac{z}{1-z}\right)^{-\frac{m+n}{2}} \cdot \frac{m}{n(1-z)^2}$$

$$= \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} z^{\frac{n}{2}-1} (1-z)^{\frac{m}{2}-1} \sim \text{Beta}\left(\frac{n}{2}, \frac{m}{2}\right)$$

$$11. C(\bar{X} - \mu_1) \sim N(0, \frac{C^2 \sigma^2}{n}) \quad d(\bar{Y} - \mu_2) \sim N(0, \frac{d^2 \sigma^2}{m})$$

$$\therefore C(\bar{X} - \mu_1) + d(\bar{Y} - \mu_2) \sim N(0, \frac{C^2 \sigma^2}{n} + \frac{d^2 \sigma^2}{m})$$

$$\frac{(n-1)S_x^2}{\sigma^2} \sim \chi^2(n-1) \quad \frac{(m-1)S_y^2}{\sigma^2} \sim \chi^2(m-1)$$

$$\therefore \frac{(n+m-2)S_w^2}{\sigma^2} \sim \chi^2(n+m-2)$$

$$C(\bar{X} - \mu_1) + d(\bar{Y} - \mu_2)$$

$$\therefore t = \frac{\sqrt{\frac{C^2 \sigma^2}{n} + \frac{d^2 \sigma^2}{m}}}{\sqrt{\frac{(n+m-2)S_w^2}{\sigma^2}}} \sim t(n+m-2)$$

(9. 设 $Y_i = F(X_i)$ 则 $F_Y(y) = P(F(X) \leq y) = P(X \leq F^{-1}(y))$
 $= y \quad y \in (0, 1)$

$\therefore Y_i \sim U(0, 1)$

设 $Z_i = -\ln Y_i$, 则 $F_Z(z) = P(-\ln Y \leq z) = 1 - e^{-z} \quad (z > 0)$

$\therefore Z_i \sim \chi^2(2)$

$\therefore T \sim \chi^2(2n)$

5.5. 4

4. $T = \sum_{i=1}^n X_i \sim N(n\mu, n)$

$$\frac{P(X_1=x_1, X_2=x_2, \dots, X_n=x_n)}{P(T=t)} = \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2}}{\frac{1}{\sqrt{2\pi n}} e^{-\frac{1}{2n} (t - n\mu)^2}}$$

$$= \frac{1}{\sqrt{2\pi}^{n-1} \cdot \sqrt{\pi}} e^{-\frac{1}{2} \left(\sum_{i=1}^n x_i^2 - \frac{t^2}{n} \right)}$$

5 分

$\therefore T$ 为充分统计量

7. ~~$f(x_1, x_2, \dots, x_n, \theta) = \theta^n a^n(x_1, x_2, \dots, x_n)^{(\theta+1)}$~~

~~证明~~ 可证: $T = x_1 x_2 \dots x_n$ 为 θ 的充分统计量

9.

(1) $f(x, \theta) = \left(\prod_{i=1}^n C_{x_i+r-1}^{r-1} \right) \theta^{nr} (1-\theta)^{\sum_{i=1}^n x_i}$

可证: $T = \sum_{i=1}^n x_i$ 为 θ 的充分统计量

(2) $p(x_m) = \frac{1}{m^n} I_{\{1 \leq x_1, \dots, x_m \leq m\}}$

$\therefore x_m$ 为充分统计量

(3) $f(x, \mu, \sigma^2) = \frac{1}{(\sqrt{2\pi})^n x_1 \dots x_n \sigma^n} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (\ln x_i)^2 + \frac{\mu}{\sigma^2} \sum_{i=1}^n \ln x_i - \frac{\mu^2}{2\sigma^2} \right\}$

$\therefore T = \left(\sum_{i=1}^n (\ln x_i)^2, \sum_{i=1}^n \ln x_i \right)$ 为充分统计量

(4) $f(x, \lambda) = 2^n \left(\prod_{i=1}^n x_i \right) \cdot \lambda^n e^{-\lambda \sum_{i=1}^n x_i^2} I_{\{x_i > 0\}}$

$\therefore T = \sum_{i=1}^n x_i^2$ 为充分统计量

$$12. f(x, \theta) = \left(\frac{1}{\theta}\right)^n \cdot \mathbb{I}_{\theta \leq x_{(1)} \leq x_{(n)} \leq \theta}$$

$\therefore (x_{(1)}, x_{(n)})$ 为 θ 的充分统计量

$$19. p(X, \theta, \mu) = \left(\frac{1}{\theta}\right)^n e^{\frac{-n\bar{x} + n\mu}{\theta}} \cdot \mathbb{I}_{x_{(1)} > \mu}$$

$\therefore (\bar{x}, x_{(1)})$ 为充分统计量

6.1.

$$6. f_{X_{(1)}}(x) = \frac{3}{\theta^3} (\theta - x)^2 \quad 0 < x < \theta$$

$$f_{X_{(3)}}(x) = \frac{3}{\theta^3} x^2$$

$$\therefore E(4X_{(1)}) = 4E(X_{(1)}) = \theta$$

$$E\left(\frac{4}{3}X_{(3)}\right) = \frac{4}{3}E(X_{(3)}) = \theta$$

都为无偏估计,

$$D(4X_{(1)}) = 16D(X_{(1)}) = \frac{3}{5}\theta^2$$

$$D\left(\frac{4}{3}X_{(3)}\right) = \frac{16}{9}D(X_{(3)}) = \frac{\theta^2}{15}$$

$\therefore \frac{4}{3}X_{(3)}$ 更有效

$$(1) E y_1 = \frac{1}{n} \cdot n E |X_1 - \bar{X}| = \sqrt{\frac{n-1}{n}} \frac{\sqrt{26}}{\sqrt{n}} \quad (X_1 - \bar{X} \sim N(0, \frac{n-1}{n} \sigma^2))$$

$$E y_2 = \frac{1}{n(n-1)} \cdot n(n-1) E |X_1 - X_2| = \frac{26}{\sqrt{n}} \quad (X_1 - X_2 \sim N(0, 2\sigma^2))$$

$$\therefore C_1 = \sqrt{\frac{n-1}{2n}} \quad C_2 = \frac{\sqrt{n}}{2}$$

6.2

$$3. (1) E X = \frac{N+1}{2}, \quad N = 2EX + 1$$

$$\therefore \hat{N} = 2\bar{X} + 1$$

$$(2) EX = \frac{2}{\theta} \quad \therefore \theta = \frac{2}{EX}$$

$$\hat{\theta} = \frac{2}{\bar{X}}$$

$$4. (1) EX = \frac{1}{3} \theta \quad \therefore \hat{\theta} = 3\bar{X}$$

$$(2) EX = \frac{\theta+1}{\theta+2} \quad \therefore \hat{\theta} = \frac{1-2\bar{X}}{\bar{X}-1}$$

6.3. 6

1.

$$(1). L(\theta) = \theta \sqrt{\theta}^n (x_1 \cdots x_n)^{\sqrt{\theta} - 1}$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{2\theta} + \frac{\ln x_1 + \ln x_2 + \cdots + \ln x_n}{2\sqrt{\theta}} = 0$$

$$\text{解: } \hat{\theta} = \left(\frac{\sum_{i=1}^n \ln x_i}{n} \right)^{-2}$$

此时, $\hat{\theta}$ 为最大值点. $\therefore \hat{\theta}$ 为 θ 的最大似然估计

$$(2) L(\theta) = \theta^n c^{n\theta} (x_1 \cdots x_n)^{-(\theta+1)}$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{\theta} + n \ln c - (\ln x_1 + \cdots + \ln x_n) = 0$$

$$\hat{\theta} = \left(\frac{1}{n} \sum_{i=1}^n \ln x_i - \ln c \right)^{-1}$$

此时, $\hat{\theta}$ 为最大值点. $\therefore \hat{\theta}$ 为 θ 的最大似然估计.